Diffusion Models in Finance: A Survey

Anonymous Author(s)
Anonymous Institution(s)

ABSTRACT

KEYWORDS

diffusion models, score-based generative models, flow matching, finance, survey

1 INTRODUCTION

What's the problem? Why people care about this survey? This should be a short background and motivation.

Why diffusion, why now. Generative modeling is useful in finance because many important regimes are rare, private, or expensive to observe: market crashes, stressed liquidity, credit defaults, customer-level fraud, and realistic high-frequency order flow. Denoising diffusion probabilistic models (DDPMs) [18] and score-based generative models [48, 49] offer stable training, strong mode coverage, and flexible conditioning. Those properties make them attractive for synthetic financial data, scenario generation, forecasting, and decision support, where GAN-style instability and mode collapse are especially costly.

Contributions. We make three contributions. First, we introduce a data-first taxonomy that organizes diffusion-family finance papers by the object being modeled: financial time series, limit order books (LOBs), tabular records, and other structured financial objects. Second, we connect those data domains to common applications–generation, prediction and inference, trading and portfolio construction, risk and stress testing, and privacy. Third, we summarize open problems around benchmarks, tail fidelity, market validity, privacy, and non-stationarity.

Scope and taxonomy of this survey. We include learned diffusion, score-based, flow-matching, rectified-flow, and stochastic-interpolant models when they are applied to financial data or provide directly relevant backbones. We exclude classical financial "diffusion" models in the purely SDE/PDE sense unless they contain a learned generative sampler. Figure 1 shows the organizing principle: the data object is the primary axis; application and method family are secondary axes. This differs from generic time-series diffusion surveys [57], tabular-diffusion surveys [7], and broad synthetic-data surveys [33], because finance imposes domain-specific requirements: no-arbitrage and measure consistency, heavy tails, stylized facts, microstructure realism, and privacy/governance.

The rest of the paper follows the same taxonomy. Section 2 gives background. Sections 3–6 review the literature by data domain. Section 7 discusses cross-cutting trends, evaluation, open problems, and directions.

2 BACKGROUND

This section fixes the common vocabulary used throughout the survey. We keep the presentation deliberately compact: the goal is not to re-derive diffusion models, but to identify the design choices that become consequential once the data are prices, order books, credit tables, or arbitrage-constrained financial objects.

Algorithm 1 Score-SDE diffusion: training and sampling

Input: data x_0, optional condition c, schedule (α_t, σ_t), score/noise network.
Training
for each minibatch **do**
  Sample $t \sim U(0, T)$ and $\epsilon \sim \mathcal{N}(0, I)$.
  Form $x_t = \alpha_t x_0 + \sigma_t \epsilon$.
  Minimize $\|s_\theta(x_t, t, c) - \nabla_{x_t} \log p(x_t \mid x_0)\|_2^2$.
  Equivalently, under Gaussian paths, minimize $\|\epsilon_\theta(x_t, t, c) - \epsilon\|_2^2$.
end for
Sampling
Draw $x_T \sim \mathcal{N}(0, I)$.
for $t = T, T - \Delta t, \ldots, \Delta t$ **do**
  Draw $z \sim \mathcal{N}(0, I)$.
  $x_{t-\Delta t} = x_t - \{f_t(x_t) - g_t^2 s_\theta(x_t, t, c)\}\Delta t + g_t \sqrt{\Delta t}\, z$.
end for
return synthetic sample $\hat{x}_0$.

Denoising diffusion and score-based foundations. DDPMs define a forward noising process that gradually transforms data $x_0 \sim p_{\text{data}}$ into an approximately Gaussian variable x_T, typically through transitions $q(x_t \mid x_{t-1})$ with a prescribed noise schedule [18]. A neural network is then trained to reverse this process, often by predicting the injected noise. The practical appeal is that training is stable and uses a simple denoising objective, while sampling produces diverse draws by running the learned reverse chain from noise to data. Score-based generative models instead learn $s_\theta(x, t, c) \approx \nabla_x \log p_t(x \mid c)$ at multiple noise levels [48]. The two views meet in the score-SDE formulation [49], where a forward process

$$dx_t = f_t(x_t)\, dt + g_t\, dw_t \tag{1}$$

is sampled in reverse by replacing the unknown score with a neural estimate:

$$dx_t = \{f_t(x_t) - g_t^2 s_\theta(x_t, t, c)\}\, dt + g_t\, d\bar{w}_t. \tag{2}$$

Here c denotes optional conditioning information and $\bar{w}_t$ is the reverse-time Brownian motion. Training usually samples x_0, a noise level t, and a perturbed observation $x_t = \alpha_t x_0 + \sigma_t \epsilon$, then fits either the score or the injected noise. Algorithm 1 gives the resulting template, following standard score-SDE notation. For finance this connection is conceptually useful: stochastic processes are already the language of asset prices, rates, volatility, and risk-neutral valuation. The learned component is not a classical calibrated diffusion coefficient, but the reverse-time score or drift that transports a tractable base distribution into an empirical financial data distribution.

Flow matching, rectified flow, and stochastic interpolants. Recent work also uses continuous-time transport formulations that learn a vector field from a base distribution to the data. In its simplest form, flow matching learns an ODE

$$\frac{dx_t}{dt} = v_\theta(x_t, t, c), \qquad x_0 \sim p_0, \tag{3}$$

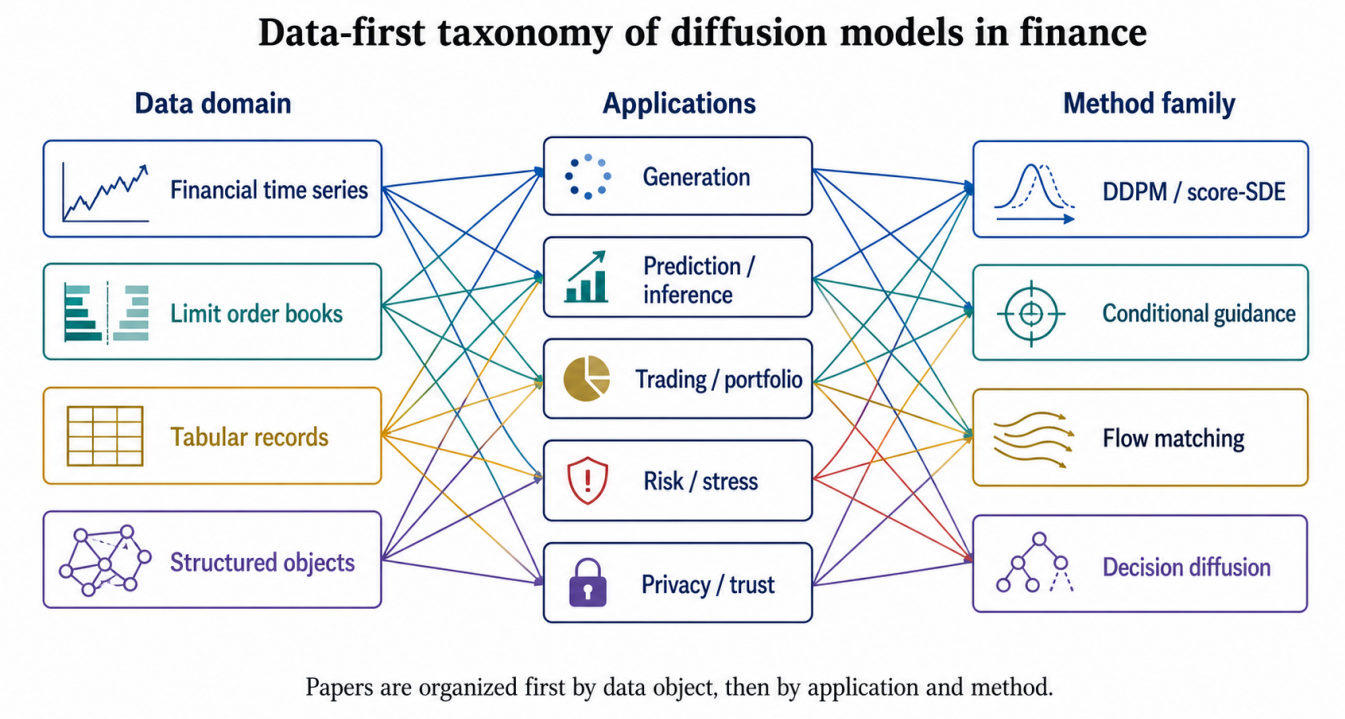


Figure 1: A data-first taxonomy for diffusion-family models in finance. Papers are organized first by the financial data object, then by application and method family.

Algorithm 2 Flow matching: training and sampling

Input: data x, base noise $z \sim p_0$, optional condition c, vector field v_θ.
Training
for each minibatch **do**
  Sample $t \sim U(0,1)$, $z \sim p_0$, and $x \sim p_{\text{data}}$.
  Set $x_t = (1-t)z + tx$ and target velocity $u_t = x - z$.
  Minimize $\|v_\theta(x_t, t, c) - u_t\|_2^2$.
end for
Sampling
Draw $x_0 \sim p_0$.
for $t = 0, \Delta t, \ldots, 1 - \Delta t$ **do**
  $x_{t+\Delta t} = x_t + \Delta t\, v_\theta(x_t, t, c)$.
end for
return synthetic sample $\hat{x}_1$.

whose terminal distribution approximates p_{data}. With a linear conditional path $x_t = (1-t)z + tx$ between noise $z \sim p_0$ and data $x \sim p_{\text{data}}$, the target velocity is $u_t = x - z$, giving the objective

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E}_{x,z,t}\left[\|v_\theta(x_t, t, c) - (x - z)\|_2^2\right]. \tag{4}$$

Algorithm 2 summarizes the corresponding training and Euler sampler [20, 36]. Rectified flow straightens the transport path to reduce sampling cost [38], and stochastic interpolants provide a broad framework that contains diffusion and flow models as special cases [2].

Conditioning and guidance. Classifier guidance modifies the reverse process using gradients from an auxiliary model, while classifier-free guidance learns conditional and unconditional scores in a single model and interpolates them at sampling time [12, 19]. Finance papers adapt this idea to regime labels, factor states, asset identities, missing-value masks, stress intensities, portfolio objectives, or structural constraints. Conditioning is therefore not a cosmetic add-on: it is the mechanism that turns a generator into a controllable market simulator, forecaster, imputer, stress-test engine, or decision module.

Sampling accelerators and architectural backbones. Sampling speed is a recurring bottleneck. DDIM [46], improved diffusion objectives [39], EDM design choices [25], and consistency models [47] seek better sample quality per function evaluation. In parallel, domain backbones determine what the model can represent: TimeGrad handles probabilistic time-series forecasting [40], CSDI addresses conditional imputation [52], SSSD combines diffusion with structured state-space sequence modeling [3], TSDiff and Diffusion-TS explore forecasting and interpretable time-series generation [27, 58], and TabDDPM/TabSyn adapt diffusion to mixed-type tabular data [28, 60]. Finance papers typically inherit these ideas, then add financial structure: asset embeddings, factor conditioning, temporal masks, privacy mechanisms, no-arbitrage penalties, or microstructure-aware encodings.

Decision-making and control. Many financial problems are not finished when a realistic sample is generated. Portfolio construction, hedging, execution, and market making require actions under uncertainty. Diffusion policies and trajectory generators provide a bridge from generative modeling to control: Diffuser models behavior trajectories for planning [23]; Decision Diffuser frames conditional generation as a decision-making primitive [1]; and the

reinforcement-learning literature surveys diffusion as policy, planner, and data generator [63]. In finance, this perspective appears when synthetic scenarios are used to train robust portfolio policies, when market generators support trading simulation, or when conditional paths feed pricing and hedging routines. The modeling question therefore shifts from "does the sample look real?" to "does the generated distribution support better decisions under the constraints of the market?"

Figure 2 summarizes the same corpus at a coarser level, showing where diffusion-family finance papers cluster and where coverage remains thin.

3 FINANCIAL TIME SERIES DATA

Financial time series form the largest cluster in the current literature. In this section, we use the term narrowly: the modeled object is an ordered sequence such as prices, returns, macroeconomic variables, etc. We therefore separate this section from LOBs in next section, where the sequence is constrained by order-book mechanics, and from structured objects such as implied-volatility surfaces or yield curves. The dominant assets are equities and ETFs, with smaller coverage of futures, energy prices, and macro-conditioned paths. Across these settings, the generator must preserve stylized facts that ordinary time-series metrics often miss, such as heavy tails, volatility clustering, leverage effects, serial dependence and cross-sectional covariance.

Two modeling choices recur. First, papers decide whether to model prices or returns. Price paths are visually and economically intuitive, but they are non-stationary and scale-sensitive; return series are closer to stationary and portfolio-ready, but harder to evaluate visually and easier to over-smooth. Second, papers decide whether the diffusion model is the final product or an intermediate object. Some works output synthetic paths for simulation or data sharing. Others use diffusion as a probabilistic forecaster, scenario tree generator, denoiser, or policy-training environment.

3.1 Unconditional Generation

Unconditional generation asks whether a model can synthesize financial time series without specifying a future condition, regime, or control signal. We read these papers along two axes: the financial variable being generated and the dimensionality of the generated object. The variable axis separates price or price-like paths from returns, return vectors, spreads, volumes, and transformed market features. The dimensionality axis separates single-instrument generation from multivariate panels that must preserve cross-asset dependence.

Most unconditional finance papers in the current corpus are stock-market papers. Takahashi and Mizuno use US intraday stock data and generate a single-stock, multivariate-feature time series: price log returns, spreads, and trading volumes are converted to wavelet images and reconstructed as synthetic market series [50]. Lesniewski and Trigila use US equity data, but the generated object is not a price path; it is a multivariate stock-return scenario used for covariance regularization and portfolio-level simulation [32]. Diffusion Factor Models move to high-dimensional stock-return panels, generating multivariate asset returns under a latent factor structure rather than independent univariate series [8]. Kim et al. use historical stock data and generate price-path-oriented series with a GBM-informed noising process, making the model closer to price dynamics than return-only scenario generation [26]. Fiaingen is useful as a stock-data-quality reference point: it works with daily stock closing-price windows and evaluates whether generated stock time series support downstream learning, although it is better treated as a synthetic-data comparator than as a core diffusion architecture [41].

3.2 Conditional Generation

Conditional generation asks whether a model can synthesize financial time series under an observed or user-specified condition. Compared with unconditional generation, the key question is not only what variable is generated, but also what information controls the generation. We read this literature along two axes: the generated data object, such as prices, returns, charts, or scenario paths, and the conditioning signal, such as trend, volatility, partial observations, stress regimes, or optimization states. This distinction is important because conditional generators are usually evaluated by whether they respond correctly to the condition, not only by whether their samples look realistic on average.

For stock data, conditional generation mainly appears as controllable stock time-series synthesis and stress-scenario construction. CoFinDiff uses Japanese stock price series and generates multivariate stock time series conditioned on user-specified trend and volatility controls, so the generated object is price-like stock series rather than return-only scenarios [51]. Diffusion-Augmented Reinforcement Learning uses stock-market return scenarios conditioned toward crash or stress regimes; the generated data are multivariate stock-return paths used to train robust portfolio policies rather than generic synthetic prices [10]. Guo et al. [17] uses financial time-series observations under partial or compressed measurements and reconstructs plausible stock and macro time-series signals, making the generated object a conditionally completed financial series rather than an unconditional market sample.

Outside stocks, conditional generation becomes more heterogeneous. Chart-based diffusion forecasting represents financial time series as visual chart objects and generates future chart continuations, with the experiments centered on Bitcoin futures price charts rather than equity panels [31]. Scenario-tree generation uses multivariate energy price time series and generates conditional scenario paths for multistage stochastic optimization, so the data object is an optimization-ready scenario tree rather than a simple price sequence [59]. These papers show why conditional generation should be separated from unconditional synthesis: the generated data must be realistic, but it must also be coherent with the conditioning signal and useful for the downstream decision problem.

3.3 Prediction and Trading

Prediction-oriented papers use diffusion models to forecast future financial states or to construct signals that feed trading and allocation decisions. In this subsection, prediction includes point or distributional forecasts of prices, returns, chart movements, denoised signals, portfolio weights, and trading actions. This boundary is useful because finance rarely evaluates prediction in isolation: a forecast is valuable only if it improves trading accuracy, risk-adjusted return,

Table 1: Representative diffusion-family finance papers organized by data domain. The table is illustrative rather than exhaustive; entries are selected to show the taxonomy used in this survey.

Paper	Year	Data domain	Region / Asset / Data	Application	Method family	Code
Beyond Monte Carlo [32]	2024	Time series	US / Stock / Return vectors	Market simulation	Diffusion	–
A Financial Time Series Denoiser [54]	2024	Time series	US / Stock / Price series	Denoising / trading signal	Conditional diffusion	–
CoFinDiff [51]	2025	Time series	JP / Stock / Price series	Controllable generation	Conditional diffusion	–
Diffusion Factor Models [8]	2025	Time series	US / Stock / Return series	Factor-structured returns	Diffusion	Yes
TRADES [6]	2025	LOB	US / Stock / LOB orderflow	Market simulation	Diffusion	Yes
DiffVolume [55]	2025	LOB	US / Stock / LOB volume	Volume generation	Diffusion	–
DiffLOB [56]	2026	LOB	US / Stock / LOB snapshots	Counterfactual generation	Diffusion	Yes
FinDiff [42]	2023	Tabular	Multi / Credit, Payment, Fund / Tabular	Tabular synthesis	Diffusion	Yes
Imb-FinDiff [44]	2024	Tabular	Multi / Finance / Tabular	Imbalance synthesis	Conditional diffusion	–
DP-FinDiff [45]	2025	Tabular	Multi / Finance / Tabular	Private synthesis	DP diffusion	Yes
IV-surface diffusion [24]	2025	Structured	US / Option / IV surface	Surface forecasting	Diffusion	Yes
Risk-neutral derivative pricing [53]	2026	Structured	Global / Stock / Return series	Derivative pricing	Diffusion	–

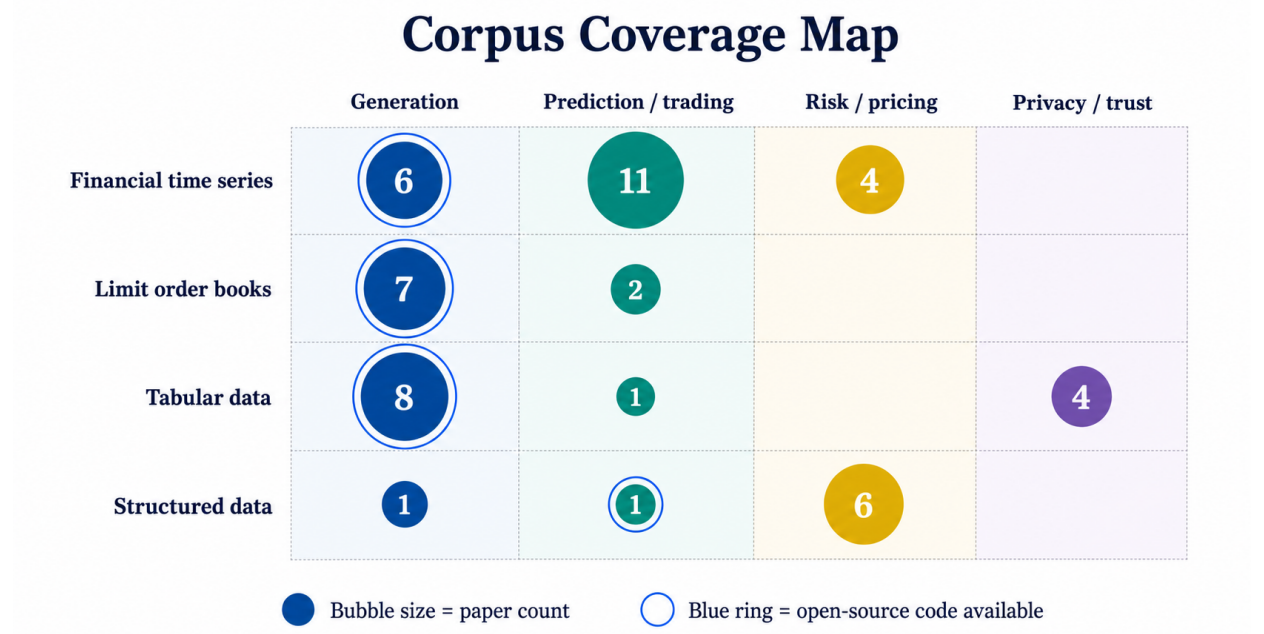


Figure 2: Corpus coverage map by financial data object and application. Cell numbers count finance-facing corpus entries; bubble size encodes paper count, and blue rings indicate cells with public implementations. Cross-listed papers may appear in multiple cells.

drawdown control, or portfolio robustness. Thus the central question is not only whether the generated distribution is realistic, but whether it is decision-useful under chronological splits, transaction costs, and changing market regimes.

Stock-level prediction papers mostly use diffusion as a way to denoise, augment, or represent financial time-series signals. Wang and Ventre [54] use US stock price series and apply a diffusion denoiser to financial time-series signals before downstream prediction and trading evaluation. Daiya et al. [11] use US stock relational time-series data and generate probabilistic stock-market predictions, making the modeled object a cross-sectional prediction distribution rather than a standalone synthetic series. Gao et al. [16] use Chinese stock factor and price-related data, with diffusion-based factor augmentation serving the prediction model rather than general market simulation. Lee et al. [31] recast financial forecasting as conditional image generation: historical price charts are treated as visual inputs, and a text-to-image diffusion model generates the next chart image to predict future price trends. These papers differ in representation, but they share the same evaluation burden: the diffusion component must improve prediction or trading signals beyond a strong chronological baseline.

Portfolio and trading papers move from forecasting signals to decision variables. Cho et al. [9] model multivariate financial time series and generate predictive return distributions for portfolio construction, so the relevant data object is a multivariate return distribution rather than a single price path. Gao et al. [14] condition stock-return generation on asset-specific factors and use the resulting samples for constrained mean-variance and mean-CVaR portfolio optimization. Alzahrani [4] focus on regime-aware and tail-risk-sensitive return scenarios for CVaR-constrained portfolio decisions, making the generated distribution explicitly risk-driven. Choudhary et al. [10] generate stress-oriented stock-return scenarios to train robust portfolio reinforcement-learning policies. Li et al. [35] shift the prediction-decision link to high-frequency trading,

where flow matching is used inside an imitation-learning policy under diverse market states. Across these works, the strongest evidence is not sample realism alone, but whether the diffusion-generated forecast, scenario, or policy improves realized financial decisions after costs, constraints, and regime changes are considered.

4 LIMIT ORDER BOOK DATA

Limit order book (LOB) data describe the visible supply and demand around the best bid and ask through price levels, queue sizes, order arrivals, cancellations, and trades. Unlike daily financial time series, LOB data are event-driven, high-dimensional, and constrained by market mechanics: prices must respect tick sizes, bid prices must remain below ask prices, queue sizes are nonnegative, and order flow must evolve in a temporally coherent way. These constraints make LOB generation more difficult than generic multivariate time-series synthesis. A useful split is therefore between papers that build generative market simulators and papers that use diffusion models for LOB prediction or forecasting.

4.1 Generative Market Simulation

Generative LOB papers aim to synthesize realistic order-book states, event streams, or market responses. Berti et al. [6] use US stock LOB order-flow data and generate realistic event-level market simulations with a diffusion model, making the data object an order-flow trajectory rather than a daily price series. Wang and Ventre [55] focus on US stock LOB volume and generate conditional volume profiles inside the order book, so the modeled variable is queue volume rather than the full event stream. Backhouse et al. [5] represent US stock LOB snapshots as image-like structures and use diffusion for market simulation, with the generated object being book states over time. ByteGen works at the orderbook-event level but uses a tokenizer-free byte-space generator, making it useful context for event-stream generation even though its architecture is not a standard diffusion model [34]. Huang et al. [21] generate controllable Chinese stock-market order flow with a diffusion-guided meta-agent, placing the emphasis on market-state control and agent-based simulation. Wang and Ventre [56] use US stock LOB snapshots and generate counterfactual order-book trajectories under regime controls, so the central object is a controllable LOB state path rather than a point forecast.

4.2 Prediction and Forecasting

LOB prediction papers use diffusion models to infer future event streams or book states from recent market history. Zheng et al. [62] provide the clearest prediction-focused example: they use US and Chinese stock LOB order-flow data and model the joint distribution of future event types and event times, making the generated object a future LOB event stream. The prediction target is therefore not a synthetic market for offline simulation, but the next sequence of high-frequency order-book events. Backhouse et al. [5] also connect LOB generation to forecasting by treating order-book states as structured image-like data and evaluating future book reconstruction. These papers suggest that diffusion-based LOB prediction should be evaluated differently from daily return forecasting: beyond directional accuracy, it must preserve event timing, queue dynamics, price-level constraints, and the short-horizon structure that trading systems actually consume.

5 TABULAR DATA

Financial tabular data consist of row-wise records such as credit applications, payment transactions, customer attributes, fund characteristics, fraud labels, and account-level features. Unlike financial time series or LOBs, these data are usually mixed-type: continuous fields, categorical variables, binary indicators, timestamps, labels, and entity identifiers may appear in the same table. The main modeling challenge is therefore not temporal coherence, but preserving feature-label relationships, categorical distributions, business constraints, rare classes, and privacy. We organize this section by the motivation for generation: synthesis and augmentation for utility, and privacy and trustworthiness for safe deployment.

5.1 Synthesis and Augmentation

The first group of tabular papers uses diffusion or flow models to generate synthetic financial records that can replace, supplement, or rebalance real training data. Sattarov et al. [42] propose FinDiff for financial tabular data generation, targeting mixed financial records such as credit, payment, and fund-like tables. The generated object is a synthetic row with heterogeneous features, so evaluation must check both marginal feature realism and joint feature-label consistency. Entity-based financial tabular synthesis extends this idea to entity-structured records, where generated rows must remain plausible within the attributes and relationships of financial entities [37]. Sattarov et al. [44] focus on class imbalance: their conditional diffusion model generates minority-class financial records so that downstream classifiers can learn rare but important events. Flow-based latent-space augmentation work follows the same utility-driven logic, using latent diffusion/flow samples to improve prediction on credit and customer tabular tasks [22]. Kuo and Motsch [30] apply a diffusion transformer to fraud-detection tables, where the generated records are useful only if they improve rare-event detection without distorting the fraud/non-fraud boundary.

5.2 Privacy and Trustworthiness

The second group treats synthetic tabular data as a privacy- and trust-constrained object. Financial institutions often cannot share raw customer, transaction, or credit records, so synthetic data are valuable only if they retain utility while reducing memorization and disclosure risk. Federated diffusion with differential privacy addresses this setting by learning tabular generators across decentralized financial datasets while limiting information leakage from any participant [43]. Privacy-preserving mixed-type diffusion models make the same concern explicit for heterogeneous financial tables, where leakage may occur through rare categorical combinations as well as continuous outliers [45]. Measuring privacy risks and tradeoffs in financial synthetic data generation shifts the emphasis from model design to evaluation: utility, fidelity, and disclosure risk must be reported together rather than optimized separately [64]. Across these papers, the core benchmark is a privacy-utility-trust triangle: generated tables should help downstream models, avoid

leaking sensitive records, and satisfy the structural rules that make financial records credible.

6 OTHER STRUCTURED FINANCIAL DATA

Not all financial objects are ordinary time series, LOB states, or tabular rows. Some data objects carry mathematical structure that must be preserved by the generator. Examples include correlation matrices, implied-volatility surfaces, yield curves, derivative paths, and risk-neutral pricing objects. These objects are structured because validity is partly defined by financial constraints: correlation matrices should be symmetric and positive semidefinite, volatility surfaces should respect no-arbitrage and cross-maturity consistency, yield curves should remain smooth and economically plausible, and derivative paths may need to respect payoff or risk-neutral dynamics. For this reason, evaluation in this section is less about generic sample realism and more about whether generated objects remain financially admissible.

Matrix, surface, and curve papers make this constraint issue explicit. Kubiak et al. [29] generate realistic financial correlation matrices with a DDPM, so the modeled object is not a price or return sequence but a structured dependence matrix used in portfolio and risk applications. Jin and Agarwal [24] forecast implied-volatility surfaces with generative diffusion models, where the generated object is a surface over strike and maturity rather than a one-dimensional volatility series. Term-structure work uses diffusion models to generate yield curves across maturities, making the data object a cross-maturity curve whose quality depends on smoothness, level-slope- curvature structure, and economic plausibility [13]. Across these papers, the challenge is that a generated object can look plausible under distributional metrics while still violating the financial structure that makes it useful.

Derivative-path and pricing papers use diffusion models for valuation-oriented structured data. Zhao and Shen [61] apply conditional diffusion to exotic option valuation and counterparty games, where generated paths must remain consistent with payoff conditions and valuation objectives. Gao et al. [15] generate solution paths of Markovian stochastic differential equations using diffusion models, placing the emphasis on path generation under specified stochastic dynamics rather than empirical market-data synthesis. Risk-neutral derivative-pricing work uses generative diffusion to approximate pricing distributions under risk-neutral measures, so the generated object is tied to valuation rather than ordinary historical forecasting [53]. These papers are best separated from generic financial time-series generation because their success depends on pricing consistency, path constraints, and financial admissibility, not only on temporal realism.

7 DISCUSSION AND CONCLUSION

Across the reviewed literature, the central pattern is not simply that diffusion models can be applied to finance, but that financial generation is becoming data-object specific. A return panel, a limit order book, a credit table, an implied-volatility surface, and a risk-neutral path are not interchangeable arrays. Each carries its own notion of validity: time-series generators must preserve tails, volatility clustering, co-movement, and regime shifts; LOB generators must respect event timing, queue dynamics, tick-size rules, and market response; tabular generators must preserve feature-label relationships while controlling privacy and rare-class behavior; structured-data generators must respect positive semidefiniteness, no-arbitrage restrictions, curve smoothness, or valuation consistency. This data-first view explains why model families alone are an insufficient taxonomy. The same denoising or flow-matching backbone can be credible in one domain and unusable in another if it violates the financial structure of the object being generated.

The most urgent open problem is benchmark and evaluation discipline. Diffusion models in image, video, and language generation became cumulative partly because the community built shared datasets, standard protocols, visible leaderboards, and increasingly demanding evaluation suites. Finance does not yet have an equivalent. Many papers use proprietary data, incompatible horizons, different preprocessing choices, weak or non-overlapping baselines, and downstream tasks that are difficult to compare. As a result, the field risks producing many plausible demonstrations but little cumulative evidence. A useful research agenda is to build benchmarks by financial data object: time-series benchmarks should test tails, volatility clustering, cross-asset dependence, regime robustness, and portfolio utility; LOB benchmarks should test event timing, queue consistency, price-level validity, and response to market actions; tabular benchmarks should jointly report fidelity, privacy leakage, constraint satisfaction, and minority-class utility; structured-data benchmarks should test positive semidefiniteness, no-arbitrage, term-structure coherence, and pricing consistency. Such benchmarks should also include chronological splits, transaction costs, stress periods, and ablations against strong non-diffusion models, so that improvement means financial usefulness rather than only sample realism.

A second open direction is scaling. In images, video, and language, diffusion models improved through larger corpora, stronger backbones, conditioning, guidance, representation learning, distillation, and faster samplers. Finance has not yet had an analogous scaling moment. The obstacles are real: high-quality financial data are proprietary, fragmented across venues and institutions, nonstationary across regimes, constrained by regulation, and expensive to label or release. Yet the opportunity is equally large. Scaled financial generators could simulate rare regimes, stress portfolios, augment scarce credit and fraud events, generate coherent order-book worlds, price structured products, and serve as environment models for trading and risk agents. The next phase of diffusion models in finance should therefore move beyond asking whether diffusion can generate financial data, and instead ask whether benchmarked, scaled, and constraint-aware generators can become trustworthy financial simulators and decision engines.

This survey argues for a data-first view of diffusion models in finance. Progress will not come from treating finance as another generic data modality or from comparing model families in isolation. It will come from building benchmarked, scaled, and constraint-aware generators whose validity is defined by the financial object they model. The next phase of diffusion models in finance should therefore move beyond asking whether diffusion can generate financial data, and instead ask whether such models can become trustworthy financial simulators and decision engines.

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